

NIZWA, Oman

WMO: 412550

Lat: 22.8591N

Lon: 57.5463E

Elev: 462

StdP: 95.90

Time Zone: 4.00 (E04)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.1	12.7	-8.9	1.9	30.4	-6.8	2.2	30.5	8.0	25.1	6.7	23.1	0.6	30	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	13.0	44.8	20.2	43.8	20.1	42.8	20.1	24.5	33.3	24.1	33.4	23.6	33.3	3.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	18.0	28.6	21.6	17.2	28.6	21.0	16.5	28.5	77.1	33.4	75.2	33.2	73.6	33.4	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	6.4	5.4	DB	7.7	47.1	1.8	0.4	6.4	47.4	5.3	47.7	4.3	47.9	2.9	48.2	
			WB	3.6	25.4	1.7	0.5	2.3	25.7	1.3	26.0	0.4	26.3	-0.9	26.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.7	19.4	21.5	25.4	30.0	34.1	35.9	35.9	34.6	32.7	29.2	24.6	20.8
	DBStd	6.23	1.98	2.54	2.34	2.38	1.86	1.97	2.34	2.29	1.86	1.76	1.87	1.94
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	22	12	5	0	0	0	0	0	0	0	0	0	4
	CDD10.0	6829	292	322	477	602	747	776	803	762	681	597	438	334
	CDD18.3	3810	46	94	218	352	488	526	544	503	431	338	188	81
	CDH23.3	57278	351	903	2620	5072	8094	8981	9222	8039	6651	4713	1952	680
Wind	WSAvg	2.1	1.6	2.0	2.2	2.5	2.6	2.6	2.6	2.3	2.2	1.9	1.6	1.6
	PrecAvg	175	11	19	26	19	12	9	25	25	10	6	6	5
	PrecMax	498	78	108	291	90	63	40	102	51	30	55	44	28
	PrecMin	72	0	5	0	0	0	0	7	2	0	0	0	0
	PrecStd	90	20	23	56	24	13	10	23	15	8	12	11	7
	DB	29.7	33.1	36.9	40.0	44.6	46.1	46.3	44.8	43.0	38.9	34.3	30.7	
	MCWB	14.5	15.7	16.1	18.0	19.7	19.7	20.4	20.6	19.6	17.9	17.0	14.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	27.9	31.2	35.4	38.9	42.6	44.8	44.8	43.5	41.4	38.0	32.9	29.2	
	2% MCWB	14.9	15.3	15.9	17.6	19.1	19.7	20.6	20.6	19.4	17.6	16.6	14.8	
	5% DB	26.6	29.7	33.9	38.0	41.4	43.8	43.6	42.4	40.4	36.9	31.9	28.1	
	5% MCWB	14.7	15.2	15.8	17.3	19.0	19.4	20.7	20.7	19.5	17.7	16.8	14.9	
	10% DB	25.4	28.3	32.5	36.9	40.3	42.5	42.5	41.2	39.2	35.8	30.8	26.9	
	10% MCWB	14.4	14.9	15.5	17.0	18.8	19.4	20.7	21.0	19.6	17.3	16.6	15.3	
	WB	17.7	19.6	20.7	22.1	24.2	25.2	24.7	24.7	24.2	22.8	21.7	19.2	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	23.4	25.2	25.4	27.6	31.6	32.7	35.3	33.8	33.1	28.7	25.8	23.0	
	2% WB	17.1	18.4	19.8	21.3	23.5	24.7	24.2	24.2	23.5	21.8	20.6	18.5	
	2% MCDB	22.8	24.9	25.6	27.7	31.3	32.3	35.0	33.9	32.4	28.7	26.1	23.3	
	5% WB	16.5	17.6	18.8	20.6	22.9	24.1	23.8	23.7	22.9	21.0	19.6	17.7	
	5% MCDB	22.6	24.6	26.0	28.4	31.5	32.2	34.6	33.8	31.7	29.1	26.1	23.2	
	10% WB	15.8	16.8	17.8	19.7	22.1	23.6	23.4	23.2	22.4	20.3	18.8	17.0	
	10% MCDB	22.5	24.2	26.8	29.7	32.1	32.4	34.1	33.6	31.4	29.2	25.8	23.0	
Mean Daily Temperature Range	MDBR	12.5	13.0	13.7	12.6	12.6	13.0	12.6	12.9	13.1	13.2	12.1	12.6	
	5% DB	13.7	14.8	15.3	13.5	13.6	14.2	13.6	13.8	13.9	13.9	13.1	14.4	
	MCWBR	4.8	5.0	4.7	3.8	4.0	4.1	3.4	3.1	3.7	3.7	4.1	4.9	
	MCDBR	11.1	11.4	10.6	10.9	12.3	12.7	12.5	12.7	13.0	11.7	10.6	10.4	
	MCWBR	4.1	4.2	4.4	4.2	4.6	4.1	2.9	2.7	3.2	3.7	4.0	4.0	
Clear-Sky Solar Irradiance	taub	0.394	0.439	0.504	0.569	0.607	0.686	0.769	0.718	0.599	0.469	0.420	0.392	
	taud	2.221	2.059	1.872	1.759	1.686	1.568	1.474	1.545	1.726	1.995	2.151	2.219	
	Ebn at Noon	862	843	805	760	727	666	614	645	717	802	825	847	
	Edh at Noon	128	161	203	231	247	276	303	282	231	169	137	124	
All-Sky Solar Radiation	RadAvg	4.75	5.54	6.40	7.06	7.62	7.41	6.86	6.79	6.47	5.91	4.95	4.50	
	RadStd	0.19	0.23	0.33	0.35	0.14	0.27	0.28	0.16	0.12	0.16	0.12	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	+0.23	N/A	N/A	N/A	+0.39	+0.45	N/A	N/A	+96	+98	

Nomenclature: See separate page